ReviewRadar : A Hybrid Deep Learning Architecture for Fake Review Detection Combining Textual and Sentiment Features

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Abstract

Usage of Online consumer reviews are a strong determinants for the purchase decision and business reputation in the digital world. But the increasing presence of fake or deceptive reviews have diminished trust on online review systems, motivating effective automated detection mechanism. This research focuses on the problem of fake reviews detection using deep learning–based text classification models. All work was tested on two benchmark datasets: Deceptive Opinion Spam and Fake Reviews dataset with forty thousand reviews. We test several deep learning architectures, such as Bidirection Long Short Term Memory (BiLSTM), BiLSTM with Attention mechanism and a transformer-based model, on the review classification task. Along with these baseline models, a new Hybrid model has been proposed which employs contextual textual representations combined with sentiment-based features to enhance the detection capability. Results show that the BiLSTM and Attention-based architectures provide very competitive performance, yet the proposed Hybrid model consistently outperforms all other models across both datasets. Thanks to its tip sequence architecture, the Hybrid model improves classification accuracy over a standard sequence models with regard to the smaller deceptive review dataset, and achieves around 91–92% accuracy on the larger fake reviews dataset. The results: learning deep contextual representations fine-tuned on auxiliary sentiment information is effective in detecting deceptive reviews. We propose a new hybrid framework that enhance trust and reliability with our methods on review websites.

Keywords

Fake Review Detection; Deceptive Opinion Spam; Natural Language Processing; Deep Learning; BiLSTM; Hybrid Model; Sentiment Analysis; Text Classification

Introduction

Online reviews are super common now, and people check them all the time before buying stuff on sites like Amazon or whatever review apps they use. Customers trust these user reviews a lot, it seems, for figuring out if a

product is good or not. But with so many reviews popping up, fake ones have become a big issue too. These deceptive reviews try to trick ratings or sway opinions, which makes the whole system less reliable.

Detecting fake reviews turns out to be tricky, especially in areas like natural language processing and machine learning. Traditional ways involved pulling out features by hand from the text or user behavior, but those often miss the deeper connections in how things are said. Deep learning has changed that some, with models like BiLSTM networks that look both ways in the text, or ones with attention mechanisms that focus on key parts. They do better at classifying text, at least from what Ive seen in some papers. In this project, we looked at deep learning for spotting fake reviews on two different datasets. One is the Deceptive Opinion Spam set, and the other is this bigger Fake Reviews dataset with over forty thousand entries, which feels like a good scale to test on. We tried out several models, including plain BiLSTM, then BiLSTM with attention added, and even a transformer-based one that handles sequences differently.We also came up with a hybrid model that mixes in textual stuff with sentiment features, kind of building on what the others do but adding that extra layer. The experiments ran on both datasets, and results showed the hybrid one performed the best overall. It is not perfect, but it edges out the rest, I think, especially when you need to catch those subtle fakes. Some parts of the evaluation got a bit messy though, like balancing the data sizes.

Dataset Description

For the evaluation of the performance of the proposed models for detecting fake reviews, two publicly available datasets were utilized in this study, namely the Ott Deceptive Opinion Spam dataset and the Kaggle Fake Reviews dataset.
The Ott Deceptive Opinion Spam dataset is one of the most popular benchmark datasets for the task of deceptive review detection. The dataset is comprised of around 1600 reviews related to hotels and was prepared for the task of opinion spam detection. The dataset is balanced, containing an equal number of truthful and deceptive reviews. The truthful reviews in this dataset come from various hotel review websites. In contrast, the deceptive reviews were created by the participants, who were told to write fake reviews for this study.
Each review in this dataset is either labeled as truthful or deceptive. Although this dataset is one of the most popular and has been utilized in various studies for the task of deceptive review detection, this dataset is not free from limitations and may affect the performance of deep learning models due to its smaller size and the fact that most of the reviews in this dataset are related to hotels.

To overcome the drawback of less data in the dataset, the Kaggle dataset was also used in this research. The Kaggle dataset contains more than 40,000 reviews in various product categories. In this dataset, each review contains text information, rating, and whether it is fake or real. The use of this dataset helps in providing more diversity in the choice of words and styles of writing, which is more suitable for deep learning models. Moreover, the use of multiple product categories in the dataset helps in improving the generality of the models. The use of these datasets helps in performing comparative experiments to test the performance of the models under different conditions. The Ott dataset is used to perform experiments because it is one of the benchmark datasets used to perform experiments in detecting deceptive reviews. However, the use of the Kaggle dataset helps in performing experiments under real-world scenarios because it contains more data in comparison to the Ott dataset. Experimental results show that the models trained on the Kaggle dataset perform better in comparison to the models trained on the Ott dataset. This is because the Kaggle dataset contains more real-world scenarios in comparison to the Ott dataset.

Related Work

Author	Year	Dataset Name	Model / Method Used	Accuracy	Contribution	Limitations
Kiana Sharifpour & Salim Lahmiri	2023	Amazon Fake Review Dataset	BiGRU; BiGRU + GloVe + Rating – Sentiment Inconsistenc y	BiGRU: ~88% Accuracy; BiGRU + GloVe + Inconsistency: >90% Accuracy	Introduced rating–sentiment inconsistency feature to improve fake review detection	Feature does not improve performance for all models
Zeinab Sedighi, Hossein Ebrahimpou r-Komleh, Ayoub Bagheri, Leila Kosseim	2019	Three-Domain Fake Review Dataset	LSTM; BiLSTM; BiLSTM + Multi-Head Self-Attention	LSTM: 81.49% F-measure; BiLSTM: ~85% F-measure; BiLSTM + Attention: 87.59% F-measure	Proposed attention-based LSTM model for deceptive review detection	Dataset size is relatively small
Yafeng Ren & Yue Zhang	2016	Three-Domain Review Dataset	CNN + GRNN; CNN + GRNN + Attention	CNN + GRNN: ~84% Accuracy; CNN + GRNN + Attention: ~86% Accuracy	Introduced neural document representation combining CNN and recurrent networks	Mainly uses textual features
Xuepeng Wang, Kang Liu, Shizhu He, Jun Zhao	2016	Yelp Review Dataset	Tensor Decomposit ion Model; Baseline Method	Tensor Model: 89.2% F1-score; Baseline: 86.1% F1-score	Learned review representation s using relational information between reviewers and products	Model complexity
Ao Li, Zhou Qin, Runshi Liu, Yiqun Yang, Dong Li	2019	Xianyu Platform Dataset	Logistic Regression Baseline; Graph Convolution al Network (GAS)	Logistic Regression: 54.86% Recall (precision 90%); GCN (GAS): 71.02% Recall (precision 90%)	Graph-based spam detection capturing local and global context	Designed mainly for large-scale graph datasets
Zeinab Sedighi et al.	2019	Three-Domain Fake Review Dataset	BiLSTM + Self-Attention	Around 88% Accuracy	Applied attention mechanism to improve fake review classification	Requires labeled datasets

Proposed Work

This study proposes a hybrid deep learning framework for detecting deceptive online reviews by integrating textual representations extracted through deep neural networks with sentiment-based features. The objective of the proposed approach is to enhance the ability of machine learning systems to identify patterns commonly observed in fake reviews, such as

exaggerated language and extreme emotional polarity.

The proposed framework performs a comparative evaluation of deep learning models, including BiLSTM, BiLSTM with attention, and a hybrid architecture that combines textual features with sentiment polarity. In addition, a transformer-based model (BERT) is evaluated separately to analyze its performance across different datasets.

Figure 1. Overall workflow of the proposed fake review detection system.

The overall workflow of the proposed system consists of the following stages:

4.1 Data Collection

Two datasets were used in this study:

1. Ott Deceptive Opinion Spam Dataset

Contains approximately 1600 hotel reviews.
Reviews are labeled as either deceptive or truthful.

2. Kaggle Fake Reviews Dataset

Contains more than 40,000 reviews across different categories.
Each review is labeled as fake or genuine.
The use of two datasets allows evaluation of model performance under different data scales and vocabulary diversity.

4.2 Data Preprocessing

Before training the models, the textual data undergoes several preprocessing steps to improve model performance.

The preprocessing pipeline includes:

- Conversion of text to lowercase
- Removal of punctuation and special characters
- Stopword removal
- Tokenization
- Padding of sequences for neural network input

These steps help standardize the review text and reduce noise within the dataset.

4.3 Word Representation

The textual reviews are converted into numerical representations using word embeddings.

Two types of embeddings are utilized:

- Trainable embedding layers for LSTM-based models
- Pretrained contextual embeddings (BERT) for transformer-based classification

Word embeddings allow the model to capture semantic relationships between words.

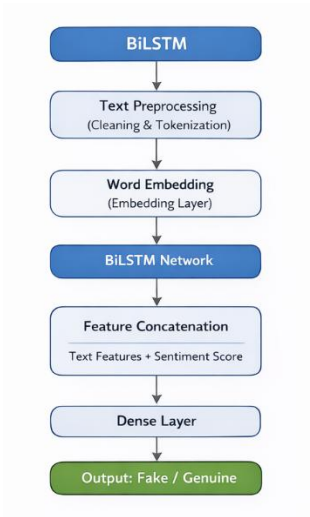
4.4 Deep Learning Models

Three deep learning architectures are evaluated for fake review detection.

4.4.1 BiLSTM Model

The Bidirectional Long Short-Term Memory (BiLSTM) network processes text sequences in both forward and backward directions, allowing the model to capture contextual dependencies within the review text.

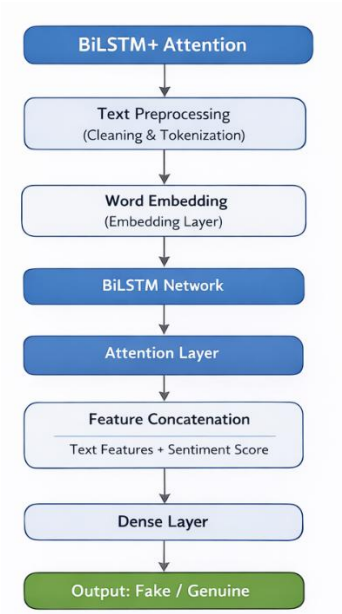
Architecture:



4.4.2 BiLSTM with Attention

To improve feature extraction, an attention mechanism is incorporated into the BiLSTM architecture. The attention layer assigns higher weights to important words within a review, enabling the model to focus on key textual patterns that may indicate deceptive content.

Architecture:



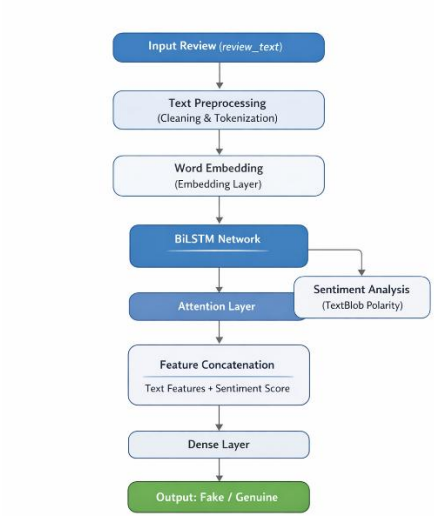
4.4.3 Proposed Hybrid Model

The proposed hybrid model integrates deep textual features with sentiment polarity features to improve classification performance. First, the review text is processed through a BiLSTM-Attention network to extract high-level textual representations. In parallel, sentiment polarity scores are computed using TextBlob. These sentiment scores capture the emotional tone of the review.

The textual feature vector and sentiment feature are then combined using a concatenation layer before being passed to a dense classification layer.

Architecture:

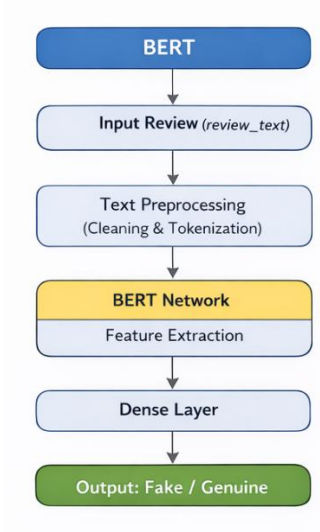
The motivation behind this hybrid approach is that fake reviews often contain exaggerated emotional language, which can be captured through sentiment analysis.



4.5 Transformer-Based Model (BERT)

To further evaluate the effectiveness of transformer architectures, the BERT model is used as a baseline for contextual text classification. BERT leverages self-attention mechanisms and pretrained language representations to capture deep semantic relationships within text.

The BERT model is fine-tuned on both datasets to analyze its performance under different dataset conditions.



Results

In this research, various deep learning models were implemented and assessed for their performance on fake review detection, including BiLSTM, BiLSTM with an attention mechanism, Hybrid Model with both textual and sentiment features, and finally, the transformer model with BERT. Experiments were carried out on two datasets: Ott's Deceptive Opinion Spam and Kaggle's Fake Reviews, which provided an opportunity to compare results on different datasets with different sizes and characteristics.

For the Kaggle data, it was found that the BiLSTM model gave an accuracy of approximately 91.0%. This shows that recurrent neural networks are capable of learning sequential patterns in text data and that the model is working correctly on fake review detection. The BiLSTM with an attention mechanism also gave similar results with an accuracy of approximately 90.9%.

This indicates that although attention helps in identifying important words in the text, it does not give significantly better results than the BiLSTM model. The proposed Hybrid Model with both textual and sentiment features gave an accuracy of approximately 91.2%. This increase in accuracy suggests that the sentiment features improve the model's ability to identify deceptive reviews, as fake reviews always contain an exaggerated amount of emotion.

Model	Accuracy	Description
BiLSTM	91.0%	Captures sequential dependencies in review text
BiLSTM + Attention	90.9%	Focuses on important words using attention mechanism
Hybrid Model	**91.2%**	Combines textual features and sentiment polarity
BERT	**93.5%**	Transformer-based contextual language model

Table 1: Model Performance on Kaggle Dataset

Model	Architecture	Key Features	Accuracy
BiLSTM	Embedding → BiLSTM → Dense	Sequential text representation	91.0%
BiLSTM + Attention	Embedding → BiLSTM → Attention → Dense	Important word weighting	90.9%
Hybrid Model	Embedding → BiLSTM → Attention + Sentiment → Dense	Textual + sentiment features	**91.2%**

Table 3: Hybrid Model vs Other Deep Learning Models

The transformer-based BERT model showed the highest level of performance compared to all other models, reaching an accuracy level of around 93.5%. This high level of performance can be attributed to the ability of the transformer-based models to understand the contextual relationships between words using the self-attention mechanism.The cross-dataset evaluation was also performed on the BERT model using the Ott dataset. The results showed that the performance level on the Ott dataset was lower compared to the Kaggle dataset. This can be attributed to the limited diversity and size of the Ott dataset, which restricts the model's ability to learn complex linguistic patterns.

The overall experimental results showed that the hybrid deep learning models perform better compared to the traditional LSTM models, while the transformer-based models showed the highest level of performance compared to all other models due to their ability to understand contextual language.

Dataset	Model	Accuracy
Ott Dataset	BERT	~82–86%
Kaggle Dataset	BERT	**93.5%**

Table 4: BERT Cross-Dataset Performance

Conclusion

In this paper, a comparative analysis of deep learning models for fake review detection was presented, with a focus on the effectiveness of hybrid models that utilize textual features along with sentiment features. Several models, including BiLSTM, BiLSTM with an attention mechanism, a hybrid deep learning model, and a transformer-based BERT model, were compared using the Ott Deceptive Opinion Spam dataset and a larger Kaggle dataset containing fake reviews.

The experimental results show that traditional deep learning models such as BiLSTM can be highly effective for detecting fake reviews. The addition of an attention mechanism enables the model to focus on the most important words within the review. The proposed hybrid model has been found to show a slight increase in the effectiveness of the model compared to traditional BiLSTM. Among all models, the best classification accuracy was obtained by the BERT model due to its transformer structure and ability to capture semantic relationships in text data. The evaluation of models across datasets also showed that using large datasets improves the performance of models, as seen by the better performance of models on the Kaggle dataset compared to the Ott dataset.

In conclusion, the results obtained in this study demonstrate the effectiveness of hybrid deep learning models for detecting deceptive online reviews and highlight the benefits of using transformer models for text classification tasks. Future research can focus on incorporating additional features such as patterns of users, graph models, and multimodal data for better fake review detection models.